

Kota Venkata Bharghav

3D Vision, Event Based Computing, Robotics

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Education

- Feb 2026 – **Doctoral Student, Marie Skłodowska-Curie doctoral fellow, ETH Zürich & University of Zürich**
Present ○ Supervisors: [Prof. Dr. Yulia Sandamirskaya](#), [Prof. Dr. Melika Payvand](#), [Prof. Dr. Fulvio Forni](#).
○ Research Topic: Event based 3D vision for safe mobile manipulation.
- Aug 2020 – **M.Sc. Chemistry & B.E. Electronics and Instrumentation, BITS Pilani, Goa, India**
July 2025 With a Minors in Data Science

Internships/Work Experience

- Aug 2025 – **Research Assistant, Peter Grünberg Institute - 15, Forschungszentrum Jülich, Germany**
Jan 2026 ○ Supervisors: [Prof. Dr. Emre Neftci](#), [Dr. Thorben Schoepe](#).
○ Built a Egomotion compensated moving object detection algorithm for SCAMP-5 SIMD vision chip
○ Worked on World Models for Grid cell-Vector Symbolic Architecture based Cognitive Localization and Mapping systems.
- Aug 2024 – **Master's Thesis, ZHAW, Switzerland**
July 2025 ○ Supervisor: [Pro. Dr. Yulia Sandamirskaya](#).
○ Thesis : *Asynchronous Graph Based Methods for Egocentric Hand Detection with Event Cameras*.
○ Also focused on hand-crafting features for Event Camera-based Simultaneous Localization and Mapping (SLAM).
- July 2023 – **Student Assistant, Brain Inspired Neural Networks Lab, BITS Goa**
July 2024 ○ Supervisor: [Prof. Dr. Basabdatta Sen Bhattacharya](#).
○ Worked on modelling the Lateral Geniculate Nucleus on SpiNNaker and contributed to the NeuCube Team in collaboration with Auckland University.
- Jan 2024 - **Semester Project, APPCAIR Lab, BITS Goa**
May 2024 ○ Supervisor : [Dr. Tanmay Tulsidas Verlekar](#)
○ Project: *Human Gait Analysis with Graph Convolutional Networks*
○ Represented gait patterns as spatio-temporal graphs, with nodes corresponding to key body joints, to analyze and classify anomalous gait patterns. Explored clinical applications by modeling on the Ataxic Gait dataset.

Peer Review Activities

Nature Communications 2025 - Early Career Researchers Reviewer

Technical Skills

Programming Languages

Python, Java, C/C++, Assembly, Bash

Deep Learning

TorchDEQ, Pytorch, Tensorflow, JAX, ONNX, CUDA

Neuromorphic Computing

Lava, Event Based Vision algorithms, SCAMP-5 vision chip, SPECK, Loihi-2, Spinnaker, NeuCube

Workshops

- March 2025 **1st Robotics Institute Germany Conference, Nuremberg, Germany**
Jan 2025 **Bangalore Neuromorphic Engineering Workshop (BNEW 2025), IISc, Bangalore**
Sept 2023 **Google - Real World ML Engineering For Content Moderation, Goa, India**

Language Skills

English IELTS Band 8, Native Fluency